arXiv:2510.04871v1 [cs.LG] 6 Oct 2025

Less is More: Recursive Reasoning with Tiny Networks

Alexia Jolicoeur-Martineau
Samsung SAIL Montréal
alexia.j@samsung.com

Abstract

Hierarchical Reasoning Model (HRM) is a novel approach using two small neural networks recursing at different frequencies. This biologically inspired method beats Large Language models (LLMs) on hard puzzle tasks such as Sudoku, Maze, and ARC-AGI while trained with small models (27M parameters) on small data ($\sim$ 1000 examples). HRM holds great promise for solving hard problems with small networks, but it is not yet well understood and may be suboptimal. We propose Tiny Recursive Model (TRM), a much simpler recursive reasoning approach that achieves significantly higher generalization than HRM, while using a single tiny network with only 2 layers. With only 7M parameters, TRM obtains 45% test-accuracy on ARC-AGI-1 and 8% on ARC-AGI-2, higher than most LLMs (e.g., Deepseek R1, o3-mini, Gemini 2.5 Pro) with less than 0.01% of the parameters.

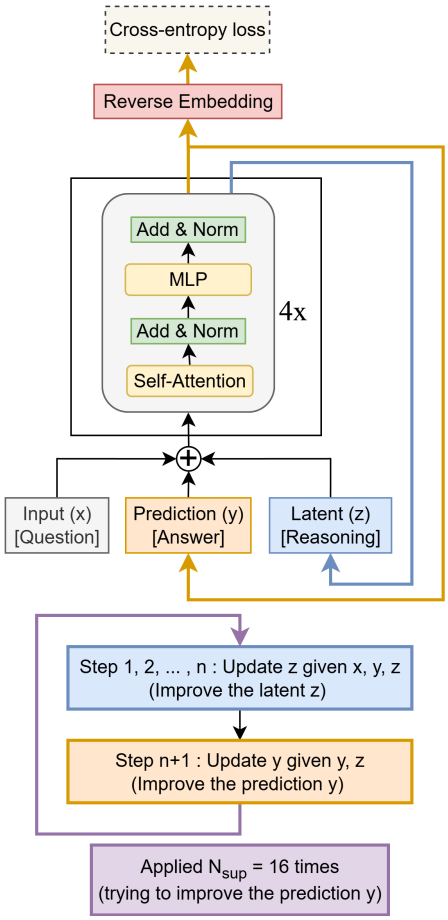


Figure 1. Tiny Recursion Model (TRM) recursively improves its predicted answer y with a tiny network. It starts with the embedded input question x and initial embedded answer y, and latent z. For up to $N_{sup} = 16$ improvements steps, it tries to improve its answer y. It does so by i) recursively updating n times its latent z given the question x, current answer y, and current latent z (recursive reasoning), and then ii) updating its answer y given the current answer y and current latent z. This recursive process allows the model to progressively improve its answer (potentially addressing any errors from its previous answer) in an extremely parameter-efficient manner while minimizing overfitting.

1. Introduction

While powerful, Large Language models (LLMs) can struggle on hard question-answer problems. Given that they generate their answer auto-regressively, there is a high risk of error since a single incorrect token can render an answer invalid. To improve their reliability, LLMs rely on Chain-of-thoughts (CoT) (Wei et al., 2022) and Test-Time Compute (TTC) (Snell et al., 2024). CoTs seek to emulate human reasoning by having the LLM to sample step-by-step reasoning traces prior to giving their answer. Doing so can improve accuracy, but CoT is expensive, requires high-quality reasoning data (which may not be available), and can be brittle since the generated reasoning may be wrong. To further improve reliability, test-time compute can be used by reporting the most common answer out of K or the highest-reward answer (Snell et al., 2024).

However, this may not be enough. LLMs with CoT and TTC are not enough to beat every problem. While LLMs have made significant progress on ARC-AGI (Chollet, 2019) since 2019, human-level accuracy still has not been reached (6 years later, as of writing of this paper). Furthermore, LLMs struggle on the newer ARC-AGI-2 (e.g., Gemini 2.5 Pro only obtains 4.9% test accuracy with a high amount of TTC) (Chollet et al., 2025; ARC Prize Foundation, 2025b).

An alternative direction has recently been proposed by Wang et al. (2025). They propose a new way forward through their novel Hierarchical Reasoning Model (HRM), which obtains high accuracy on puzzle tasks where LLMs struggle to make a dent (e.g., Sudoku solving, Maze pathfinding, and ARC-AGI). HRM is a supervised learning model with two main novelties: 1) *recursive hierarchical reasoning*, and 2) *deep supervision*.

Recursive hierarchical reasoning consists of recursing multiple times through two small networks (f_L at high frequency and f_H at low frequency) to predict the answer. Each network generates a different latent feature: f_L outputs z_H and f_H outputs z_L. Both features (z_L, z_H) are used as input to the two networks. The authors provide some biological arguments in favor of recursing at different hierarchies based on the different temporal frequencies at which the brains operate and hierarchical processing of sensory inputs.

Deep supervision consists of improving the answer through multiple supervision steps while carrying the two latent features as initialization for the improvement steps (after detaching them from the computational graph so that their gradients do not propagate). This provide residual connections, which emulates very deep neural networks that are too memory expensive to apply in one forward pass.

An independent analysis on the ARC-AGI benchmark showed that deep supervision seems to be the primary driver of the performance gains (ARC Prize Foundation, 2025a). Using *deep supervision* doubled accuracy over single-step supervision (going from 19% to 39% accuracy), while *recursive hierarchical reasoning* only slightly improved accuracy over a regular model with a single forward pass (going from 35.7% to 39.0% accuracy). This suggests that reasoning across different supervision steps is worth it, but the recursion done in each supervision step is not particularly important.

In this work, we show that the benefit from *recursive reasoning* can be massively improved, making it much more than incremental. We propose Tiny Recursive Model (TRM), an improved and simplified approach using a much smaller tiny network with only 2 layers that achieves significantly higher generalization than HRM on a variety of problems. In doing so, we improve the state-of-the-art test accuracy on Sudoku-Extreme from 55% to 87%, Maze-Hard from 75% to 85%, ARC-AGI-1 from 40% to 45%, and ARC-AGI-2 from 5% to 8%.

2. Background

HRM is described in Algorithm 2. We discuss the details of the algorithm further below.

2.1. Structure and goal

The focus of HRM is supervised learning. Given an input, produce an output. Both input and output are assumed to have shape $[B, L]$ (when the shape differs, padding tokens can be added), where B is the batch-size and L is the context-length.

HRM contains four learnable components: the input embedding $f_I(\cdot;\theta_I)$, low-level recurrent network $f_L(\cdot;\theta_L)$, high-level recurrent network $f_H(\cdot;\theta_H)$, and the output head $f_O(\cdot;\theta_O)$. Once the input is embedded, the shape becomes $[B, L, D]$ where D is the embedding size. Each network is a 4-layer Transformers architecture (Vaswani et al., 2017), with RMSNorm (Zhang & Sennrich, 2019), no bias (Chowdhery et al., 2023), rotary embeddings (Su et al., 2024), and SwiGLU activation function (Hendrycks & Gimpel, 2016; Shazeer, 2020).

2.2. Recursion at two different frequencies

Given the hyperparameters used by Wang et al. (2025) ($n = 2$ f_L steps, 1 f_H steps; done $T = 2$ times), a forward pass of HRM is done as follows:

$$
\begin{aligned}
x &\leftarrow f_I(\tilde{x}) \\
z_L &\leftarrow f_L\left(z_L + z_H + x\right) && \text{\# without gradients} \\
z_L &\leftarrow f_L\left(z_L + z_H + x\right) && \text{\# without gradients} \\
z_H &\leftarrow f_H\left(z_L + z_H\right) && \text{\# without gradients} \\
z_L &\leftarrow f_L\left(z_L + z_H + x\right) && \text{\# without gradients} \\
z_L &\leftarrow z_L.detach() \\
z_H &\leftarrow z_H.detach() \\
z_L &\leftarrow f_L\left(z_L + z_H + x\right) && \text{\# with gradients} \\
z_H &\leftarrow f_H\left(z_L + z_H\right) && \text{\# with gradients} \\
\hat{y} &\leftarrow \text{argmax}(f_O\left(z_H\right))
\end{aligned}
$$

where $\hat{y}$ is the predicted output answer, z_L and z_H are either initialized embeddings or the embeddings of the previous deep supervision step (after detaching them from the computational graph). As can be seen,

```
def hrm(z, x, n=2, T=2): # hierarchical reasoning
    zH, zL = z
    with torch.no_grad():
        for i in range(nT - 2):
            zL = L_net(zL, zH, x)
            if (i + 1) % T == 0:
                zH = H_net(zH, zL)
    # 1-step grad
    zL = L_net(zL, zH, x)
    zH = H_net(zH, zL)
    return (zH, zL), output_head(zH), Q_head(zH)

def ACT_halt(q, y_hat, y_true):
    target_halt = (y_hat == y_true)
    loss = 0.5*binary_cross_entropy(q[0], target_halt)
    return loss

def ACT_continue(q, last_step):
    if last_step:
        target_continue = sigmoid(q[0])
    else:
        target_continue = sigmoid(max(q[0], q[1]))
    loss = 0.5*binary_cross_entropy(q[1], target_continue)
    return loss

# Deep Supervision
for x_input, y_true in train_dataloader:
    z = z_init
    for step in range(N_sup): # deep supervision
        x = input_embedding(x_input)
        z, y_pred, q = hrm(z, x)
        loss = softmax_cross_entropy(y_pred, y_true)
        # Adaptive computational time (ACT) using Q-learning
        loss += ACT_halt(q, y_pred, y_true)
        _, _, q_next = hrm(z, x) # extra forward pass
        loss += ACT_continue(q_next, step == N_sup - 1)
        z = z.detach()
        loss.backward()
        opt.step()
        opt.zero_grad()
        if q[0] > q[1]: # early-stopping
            break
```

Figure 2. Pseudocode of Hierarchical Reasoning Models (HRMs).

a forward pass of HRM consists of applying 6 function evaluations, where the first 4 function evaluations are detached from the computational graph and are not back-propagated through. The authors uses $n = 2$ with $T = 2$ in all experiments, but HRM can be generalized by allowing for an arbitrary number of L steps (n) and recursions (T) as shown in Algorithm 2.

2.3. Fixed-point recursion with 1-step gradient approximation

Assuming that (z_L, z_H) reaches a fixed-point (z_L^*, z_H^*) through recursing from both f_L and f_H,

$$z_L^* \approx f_L\left(z_L^* + z_H + x\right)$$
$$z_H^* \approx f_H\left(z_L + z_H^*\right),$$

the Implicit Function Theorem (Krantz & Parks, 2002) with the 1-step gradient approximation (Bai et al., 2019) is used to approximate the gradient by back-propagating only the last f_L and f_H steps. This theorem is used to justify only tracking the gradients of the last two steps (out of 6), which greatly reduces memory demands.

2.4. Deep supervision

To improve effective depth, deep supervision is used. This consists of reusing the previous latent features (z_H and z_L) as initialization for the next forward pass. This allows the model to reason over many iterations and improve its latent features (z_L and z_H) until it (hopefully) converges to the correct solution. At most $N_{sup} = 16$ supervision steps are used.

2.5. Adaptive computational time (ACT)

With deep supervision, each mini-batch of data samples must be used for $N_{sup} = 16$ supervision steps before moving to the next mini-batch. This is expensive, and there is a balance to be reached between optimizing a few data examples for many supervision steps versus optimizing many data examples with less supervision steps. To reach a better balance, a halting mechanism is incorporated to determine whether the model should terminate early. It is learned through a Q-learning objective that requires passing the z_H through an additional head and running an additional forward pass (to determine if halting now rather than later would have been preferable). They call this method Adaptive computational time (ACT). It is only used during training, while the full $N_{sup} = 16$ supervision steps are done at test time to maximize downstream performance. ACT greatly diminishes the time spent per example (on average spending less than 2 steps on the Sudoku-Extreme dataset rather than the full $N_{sup} = 16$ steps), allowing more coverage of the dataset given a fixed number of training iterations.

2.6. Deep supervision and 1-step gradient approximations replaces BPTT

Deep supervision and the 1-step gradient approximation provide a more biologically plausible and less computationally-expansive alternative to Backpropagation Through Time (BPTT) (Werbos, 1974; Rumelhart et al., 1985; LeCun, 1985) for solving the temporal credit assignment (TCA) (Rumelhart et al., 1985; Werbos, 1988; Elman, 1990) problem (Lillicrap & Santoro, 2019). The implication is that HRM can learn what would normally require an extremely large network without having to back-propagate through its entire depth. Given the hyperparameters used by Jang et al. (2023) in all their experiments, HRM effectively reasons over $n_{layers}(n+1)TN_{sup} = 4 * (2+1) * 2 * 16 = 384$ layers of effective depth.

2.7. Summary of HRM

HRM leverages recursion from two networks at different frequencies (high frequency versus low frequency) and deep supervision to learn to improve its answer over multiple supervision steps (with ACT to reduce time spent per data example). This enables the model to imitate extremely large depth without requiring backpropagation through all layers. This approach obtains significantly higher performance on hard question-answer tasks that regular supervised models struggle with. However, this method is quite complicated, relying a bit too heavily on uncertain biological arguments and fixed-point theorems that are not guaranteed to be applicable. In the next section, we discuss those issues and potential targets for improvements in HRM.

3. Target for improvements in Hierarchical Reasoning Models

In this section, we identify key targets for improvements in HRM, which will be addressed by our proposed method, Tiny Recursion Models (TRMs).

3.1. Implicit Function Theorem (IFT) with 1-step gradient approximation

HRM only back-propagates through the last 2 of the 6 recursions. The authors justify this by leveraging the Implicit Function Theorem (IFT) and one-step approximation, which states that when a recurrent function converges to a fixed point, backpropagation can be applied in a single step at that equilibrium point.

There are concerns about applying this theorem to HRM. Most importantly, there is no guarantee that a fixed-point is reached. Deep equilibrium models normally do fixed-point iteration to solve for the fixed point$z^* = f(z^*)$ (Bai et al., 2019). However, in the case of HRM, it is not iterating to the fixed-point but simply doing forward passes of f_L and f_H. To make matters worse, HRM is only doing 4 recursions before stopping to apply the one-step approximation. After its first loop of two f_L and 1 f_H evaluations, it only apply a single f_L evaluation before assuming that a fixed-point is reached for both z_L and z_H ($z_L^* = f_L(z_L^* + z_H + x)$ and $z_H^* = f_H(z_L^* + z_H^*)$). Then, the one-step gradient approximation is applied to both latent variables in succession.

The authors justify that a fixed-point is reached by depicting an example with $n = 7$ and $T = 7$ where the forward residuals is reduced over time (Figure 3 in Wang et al. (2025)). Even in this setting, which is different from the much smaller $n = 2$ and $T = 2$ used in every experiment of their paper, we observe the following:

1. the residual for z_H is clearly well above 0 at every step
2. the residual for z_L only becomes closer to 0 after many cycles, but it remains significantly above 0
3. z_L is very far from converged after one f_L evaluation at T cycles, which is when the fixed-point is assumed to be reached and the 1-step gradient approximation is used

Thus, while the application of the IFT theorem and 1-step gradient approximation to HRM has some basis since the residuals do generally reduce over time, a fixed point is unlikely to be reached when the theorem is actually applied.

In the next section, we show that we can bypass the need for the IFT theorem and 1-step gradient approximation, thus bypassing the issue entirely.

3.2. Twice the forward passes with Adaptive computational time (ACT)

HRM uses Adaptive computational time (ACT) during training to optimize the time spent of each data sample. Without ACT, $N_{sup} = 16$ supervision steps would be spent on the same data sample, which is highly inefficient. They implement ACT through an additional Q-learning objective, which decides when to halt and move to a new data sample rather than keep iterating on the same data. This allows much more efficient use of time especially since the average number of supervision steps during training is quite low with ACT (less than 2 steps on the Sudoku-Extreme dataset as per their reported numbers).

However, ACT comes at a cost. This cost is not directly shown in the HRM's paper, but it is shown in their official code. The Q-learning objective relies on a halting loss and a continue loss. The continue loss requires an extra forward pass through HRM (with all 6 function evaluations). This means that while ACT optimizes time more efficiently per sample, it requires 2 forward passes per optimization step. The exact formulation is shown in Algorithm 2.

In the next section, we show that we can bypass the need for two forward passes in ACT.

3.3. Hierarchical interpretation based on complex biological arguments

The HRM's authors justify the two latent variables and two networks operating at different hierarchies based on biological arguments, which are very far from artificial neural networks. They even try to match HRM to actual brain experiments on mice. While interesting, this sort of explanation makes it incredibly hard to parse out why HRM is designed the way it is. Given the lack of ablation table in their paper, the over-reliance on biological arguments and fixed-point theorems (that are not perfectly applicable), it is hard to determine what parts of HRM is helping what and why. Furthermore, it is not clear why they use two latent features rather than other combinations of features.

In the next section, we show that the recursive process can be greatly simplified and understood in a much simpler manner that does not require any biological argument, fixed-point theorem, hierarchical interpretation, nor using two networks. It also explains why 2 is the optimal number of features (z_L and z_H).

```
def latent_recursion(x, y, z, n=6):
    for i in range(n): # latent reasoning
        z = net(x, y, z)
    y = net(y, z) # refine output answer
    return y, z

def deep_recursion(x, y, z, n=6, T=3):
    # recursing T-1 times to improve y and z (no gradients needed)
    with torch.no_grad():
        for j in range(T-1):
            y, z = latent_recursion(x, y, z, n)
    # recursing once to improve y and z
    y, z = latent_recursion(x, y, z, n)
    return (y.detach(), z.detach()), output_head(y), Q_head(y)

# Deep Supervision
for x_input, y_true in train_dataloader:
    y, z = y_init, z_init
    for step in range(N_supervision):
        x = input_embedding(x_input)
        (y, z), y_hat, q_hat = deep_recursion(x, y, z)
        loss = softmax_cross_entropy(y_hat, y_true)
        loss += binary_cross_entropy(q_hat, (y_hat == y_true))
        loss.backward()
        opt.step()
        opt.zero_grad()
        if q_hat > 0: # early-stopping
            break
```

Figure 3. Pseudocode of Tiny Recursion Models (TRMs).

4. Tiny Recursion Models

In this section, we present Tiny Recursion Models (TRMs). Contrary to HRM, TRM requires no complex mathematical theorem, hierarchy, nor biological arguments. It generalizes better while requiring only a single tiny network (instead of two medium-size networks) and a single forward pass for the ACT (instead of 2 passes). Our approach is described in Algorithm 3 and illustrated in Figure 1. We also provide an ablation in Table 1 on the Sudoku-Extreme dataset (a dataset of difficult Sudokus with only 1K training examples, but 423K test examples). Below, we explain the key components of TRMs.

Table 1. Ablation of TRM on Sudoku-Extreme comparing % Test accuracy, effective depth per supervision step ($T(n+1)n_{layers}$), number of Forward Passes (NFP) per optimization step, and number of parameters

Method	Acc (%)	Depth	NFP	# Params
HRM	55.0	24	2	27M
TRM ($T=3, n=6$)	87.4	42	1	5M
w/ ACT	86.1	42	2	5M
w/ separate f_H, f_L	82.4	42	1	10M
no EMA	79.9	42	1	5M
w/ 4-layers, $n=3$	79.5	48	1	10M
w/ self-attention	74.7	42	1	7M
w/ $T=2, n=2$	73.7	12	1	5M
w/ 1-step gradient	56.5	42	1	5M

4.1. No fixed-point theorem required

HRM assumes that the recursions converge to a fixed-point for both z_L and z_H in order to leverage the 1-step gradient approximation (Bai et al., 2019). This allows the authors to justify only back-propagating through the last two function evaluations (1 f_L and 1 f_H). To bypass this theoretical requirement, we define a full recursion process as containing n evaluations of f_L and 1 evaluation of f_H:

$$\begin{aligned} z_L &\leftarrow f_L\left(z_L + z_H + x\right) \\ &\dots \\ z_L &\leftarrow f_L\left(z_L + z_H + x\right) \\ z_H &\leftarrow f_H\left(z_L + z_H\right). \end{aligned}$$

Then, we simply back-propagate through the full recursion process.

Through deep supervision, the models learns to take any (z_L, z_H) and improve it through a full recursion process, hopefully making z_H closer to the solution. This means that by the design of the deep supervision goal, running a few full recursion processes (even without gradients) is expected to bring us closer to the solution. We propose to run $T-1$ recursion processes without gradient to improve (z_L, z_H) before running one recursion process with backpropagation.

Thus, instead of using the 1-step gradient approximation, we apply a full recursion process containing n evaluations of f_L and 1 evaluation of f_H. This removes entirely the need to assume that a fixed-point

is reached and the use of the IFT theorem with 1-step gradient approximation. Yet, we can still leverage multiple backpropagation-free recursion processes to improve (z_L, z_H). With this approach, we obtain a massive boost in generalization on Sudoku-Extreme (improving TRM from 56.5% to 87.4%; see Table 1).

4.2. Simpler reinterpretation of z_H and z_L

HRM is interpreted as doing hierarchical reasoning over two latent features of different hierarchies due to arguments from biology. However, one might wonder why use two latent features instead of 1, 3, or more? And do we really need to justify these so-called "hierarchical" features based on biology to make sense of them? We propose a simple non-biological explanation, which is more natural, and directly answers the question of why there are 2 features.

The fact of the matter is: z_H is simply the current (embedded) solution. The embedding is reversed by applying the output head and rounding to the nearest token using the argmax operation. On the other hand, z_L is a latent feature that does not directly correspond to a solution, but it can be transformed into a solution by applying $z_H \leftarrow f_H(x, z_L, z_H)$. We show an example on Sudoku-Extreme in Figure 6 to highlight the fact that z_H does correspond to the solution, but z_L does not.

Once this is understood, hierarchy is not needed; there is simply an input x, a proposed solution y (previously called z_H), and a latent reasoning feature z (previously called z_L). Given the input question x, current solution y, and current latent reasoning z, the model recursively improves its latent z. Then, given the current latent z and the previous solution y, the model proposes a new solution y (or stay at the current solution if its already good).

Although this has no direct influence on the algorithm, this re-interpretation is much simpler and natural. It answers the question about why two features: remembering in context the question x, previous reasoning z, and previous answer y helps the model iterate on the next reasoning z and then the next answer y. If we were not passing the previous reasoning z, the model would forget how it got to the previous solution y (since z acts similarly as a chain-of-thought). If we were not passing the previous solution y, then the model would forget what solution it had and would be forced to store the solution y within z instead of using it for latent reasoning. Thus, we need both y and z separately, and there is no apparent reason why one would need to split z into multiple features.

While this is intuitive, we wanted to verify whether using more or less features could be helpful. Results are shown in Table 2.

More features (> 2): We tested splitting z into different features by treating each of the n recursions as producing a different z_i for $i = 1, ..., n$. Then, each z_i is carried across supervision steps. The approach is described in Algorithm 5. In doing so, we found performance to drop. This is expected because, as discussed, there is no apparent need for splitting z into multiple parts. It does not have to be hierarchical.

Single feature: Similarly, we tested the idea of taking a single feature by only carrying z_H across supervision steps. The approach is described in Algorithm 4. In doing so, we found performance to drop. This is expected because, as discussed, it forces the model to store the solution y within z.

Thus, we explored using more or less latent variables on Sudoku-Extreme, but found that having only y and z lead to better test accuracy in addition to being the simplest more natural approach.

Table 2. TRM on Sudoku-Extreme comparing % Test accuracy when using more or less latent features

Method	# of features	Acc (%)
TRM y, z (Ours)	2	87.4
TRM multi-scale z	$n + 1 = 7$	77.6
TRM single z	1	71.9

4.3. Single network

HRM uses two networks, one applied frequently as a *low-level* module f_H and one applied rarely as an *high-level* module (f_H). This requires twice the number of parameters compared to regular supervised learning with a single network.

As mentioned previously, while f_L iterates on the latent reasoning feature z (z_L in HRM), the goal of f_H is to update the solution y (z_H in HRM) given the latent reasoning and current solution. Importantly, since $z \leftarrow f_L(x + y + z)$ contains x but $y \leftarrow f_H(y + z)$ does not contains x, the task to achieve (iterating on z versus using z to update y) is directly specified by the inclusion or lack of x in the inputs. Thus, we considered the possibility that both networks could be replaced by a single network doing both tasks. In doing so, we obtain better generalization on Sudoku-Extreme (improving TRM from 82.4% to 87.4%; see Table 1) while reducing the number of parameters by half. It turns out that a single network is enough.

4.4. Less is more

We attempted to increase capacity by increasing the number of layers in order to scale the model. Surprisingly, we found that adding layers decreased generalization due to overfitting. In doing the opposite, decreasing the number of layers while scaling the number of recursions (n) proportionally (to keep the amount of compute and emulated depth approximately the same), we found that using 2 layers (instead of 4 layers) maximized generalization. In doing so, we obtain better generalization on Sudoku-Extreme (improving TRM from 79.5% to 87.4%; see Table 1) while reducing the number of parameters by half (again).

It is quite surprising that smaller networks are better, but 2 layers seems to be the optimal choice. Bai & Melas-Kyriazi (2024) also observed optimal performance for 2-layers in the context of deep equilibrium diffusion models; however, they had similar performance to the bigger networks, while we instead observe better performance with 2 layers. This may appear unusual, as with modern neural networks, generalization tends to directly correlate with model sizes. However, when data is too scarce and model size is large, there can be an overfitting penalty (Kaplan et al., 2020). This is likely an indication that there is too little data. Thus, using tiny networks with deep recursion and deep supervision appears to allow us to bypass a lot of the overfitting.

4.5. attention-free architecture for tasks with small fixed context length

Self-attention is particularly good for long-context lengths when $L \gg D$ since it only requires a matrix of $[D, 3D]$ parameters, even though it can account for the whole sequence. However, when focusing on tasks where $L \leq D$, a linear layer is cheap, requiring only a matrix of $[L, L]$ parameters. Taking inspiration from the MLP-Mixer (Tolstikhin et al., 2021), we can replace the self-attention layer with a multilayer perceptron (MLP) applied on the sequence length. Using an MLP instead of self-attention, we obtain better generalization on Sudoku-Extreme (improving from 74.7% to 87.4%; see Table 1). This worked well on Sudoku 9x9 grids, given the small and fixed context length; however, we found this architecture to be suboptimal for tasks with large context length, such as Maze-Hard and ARC-AGI (both using 30x30 grids). We show results with and without self-attention for all experiments.

4.6. No additional forward pass needed with ACT

As previously mentioned, the implementation of ACT in HRM through Q-learning requires two forward passes, which slows down training. We propose a simple solution, which is to get rid of the continue loss (from the Q-learning) and only learn a halting probability through a Binary-Cross-Entropy loss of having reached the correct solution. By removing the continue loss, we remove the need for the expensive second forward pass, while still being able to determine when to halt with relatively good accuracy. We found no significant difference in generalization from this change (going from 86.1% to 87.4%; see Table 1).

4.7. Exponential Moving Average (EMA)

On small data (such as Sudoku-Extreme and Maze-Hard), HRM tends to overfit quickly and then diverge. To reduce this problem and improves stability, we integrate Exponential Moving Average (EMA) of the weights, a common technique in GANs and diffusion models to improve stability (Brock et al., 2018; Song & Ermon, 2020). We find that it prevents sharp collapse and leads to higher generalization (going from 79.9% to 87.4%; see Table 1).

4.8. Optimal the number of recursions

We experimented with different number of recursions by varying T and n and found that $T = 3, n = 3$ (equivalent to 48 recursions) in HRM and $T = 3, n = 6$ in TRM (equivalent to 42 recursions) to lead to optimal generalization on Sudoku-Extreme. More recursions could be helpful for harder problems (we have not tested it, given our limited resources); however, increasing either T or n incurs massive slowdowns. We show results at different n and T for HRM and TRM in Table 3. Note that TRM requires backpropagation through a full recursion process, thus increasing n too much leads to Out Of Memory (OOM) errors. However, this memory cost is well worth its price in gold.

In the following section, we show our main results on multiple datasets comparing HRM, TRM, and LLMs.

5. Results

Following Wang et al. (2025), we test our approach on the following datasets: Sudoku-Extreme (Wang et al., 2025), Maze-Hard (Wang et al., 2025), ARC-AGI-1 (Chollet, 2019) and, ARC-AGI-2 (Chollet et al., 2025). Results are presented in Tables 4 and 5. Hyperparameters are detailed in Section 6. Datasets are discussed below.

Table 3. % Test accuracy on Sudoku-Extreme dataset. HRM versus TRM matched at a similar effective depth per supervision step ($T(n+1)n_{layers}$)

		HRM $n=k$, 4 layers		TRM $n=2k$, 2 layers	
k	T	Depth	Acc (%)	Depth	Acc (%)
1	1	9	46.4	7	63.2
2	2	24	55.0	20	81.9
3	3	48	61.6	42	87.4
4	4	80	59.5	72	84.2
6	3	84	62.3	78	OOM
3	6	96	58.8	84	85.8
6	6	168	57.5	156	OOM

Sudoku-Extreme consists of extremely difficult Sudoku puzzles (Dillion, 2025; Palm et al., 2018; Park, 2018) (9x9 grid), for which only 1K training samples are used to test small-sample learning. Testing is done on 423K samples. Maze-Hard consists of 30x30 mazes generated by the procedure by Lehnert et al. (2024) whose shortest path is of length above 110; both the training set and test set include 1000 mazes.

ARC-AGI-1 and ARC-AGI-2 are geometric puzzles involving monetary prizes. Each puzzle is designed to be easy for a human, yet hard for current AI models. Each puzzle task consists of 2-3 input–output demonstration pairs and 1-2 test inputs to be solved. The final score is computed as the accuracy over all test inputs from two attempts to produce the correct output grid. The maximum grid size is 30x30. ARC-AGI-1 contains 800 tasks, while ARC-AGI-2 contains 1120 tasks. We also augment our data with the 160 tasks from the closely related ConceptARC dataset (Moskvichev et al., 2023). We provide results on the public evaluation set for both ARC-AGI-1 and ARC-AGI-2.

While these datasets are small, heavy data-augmentation is used in order to improve generalization. Sudoku-Extreme uses 1000 shuffling (done without breaking the Sudoku rules) augmentations per data example. Maze-Hard uses 8 dihedral transformations per data example. ARC-AGI uses 1000 data augmentations (color permutation, dihedral-group, and translations transformations) per data example. The dihedral-group transformations consist of random 90-degree rotations, horizontal/vertical flips, and reflections.

From the results, we see that TRM without self-attention obtains the best generalization on Sudoku-Extreme (87.4% test accuracy). Meanwhile, TRM with self-attention generalizes better on the other tasks (probably due to inductive biases and the overcapacity of the MLP on large 30x30 grids). TRM with self-attention obtains 85.3% accuracy on Maze-Hard, 44.6% accuracy on ARC-AGI-1, and 7.8% accuracy on ARC-AGI-2 with 7M parameters. This is significantly higher than the 74.5%, 40.3%, and 5.0% obtained by HRM using 4 times the number of parameters (27M).

Table 4. % Test accuracy on Puzzle Benchmarks (Sudoku-Extreme and Maze-Hard)

Method	# Params	Sudoku	Maze
Chain-of-thought, pretrained			
Deepseek R1	671B	0.0	0.0
Claude 3.7 8K	?	0.0	0.0
O3-mini-high	?	0.0	0.0
Direct prediction, small-sample training			
Direct pred	27M	0.0	0.0
HRM	27M	55.0	74.5
TRM-Att (Ours)	7M	74.7	85.3
TRM-MLP (Ours)	5M/19M[1]	87.4	0.0

Table 5. % Test accuracy on ARC-AGI Benchmarks (2 tries)

Method	# Params	ARC-1	ARC-2
Chain-of-thought, pretrained			
Deepseek R1	671B	15.8	1.3
Claude 3.7 16K	?	28.6	0.7
o3-mini-high	?	34.5	3.0
Gemini 2.5 Pro 32K	?	37.0	4.9
Grok-4-thinking	1.7T	66.7	16.0
Bespoke (Grok-4)	1.7T	79.6	29.4
Direct prediction, small-sample training			
Direct pred	27M	21.0	0.0
HRM	27M	40.3	5.0
TRM-Att (Ours)	7M	44.6	7.8
TRM-MLP (Ours)	19M	29.6	2.4

6. Conclusion

We propose Tiny Recursion Models (TRM), a simple recursive reasoning approach that achieves strong generalization on hard tasks using a single tiny network recursing on its latent reasoning feature and progressively improving its final answer. Contrary to the Hierarchical Reasoning Model (HRM), TRM requires no fixed-point theorem, no complex biological justifications, and no hierarchy. It significantly reduces the number of parameters by halving the number of layers and replacing the two networks with a single tiny network. It also simplifies the halting process, removing the need for the extra forward pass. Over-

[1] 5M on Sudoku and 19M on Maze

all, TRM is much simpler than HRM, while achieving better generalization.

While our approach led to better generalization on 4 benchmarks, every choice made is not guaranteed to be optimal on every dataset. For example, we found that replacing the self-attention with an MLP worked extremely well on Sudoku-Extreme (improving test accuracy by 10%), but poorly on other datasets. Different problem settings may require different architectures or number of parameters. Scaling laws are needed to parametrize these networks optimally. Although we simplified and improved on deep recursion, the question of why recursion helps so much compared to using a larger and deeper network remains to be explained; we suspect it has to do with overfitting, but we have no theory to back this explaination. Not all our ideas made the cut; we briefly discuss some of the failed ideas that we tried but did not work in Section 6. Currently, recursive reasoning models such as HRM and TRM are supervised learning methods rather than generative models. This means that given an input question, they can only provide a single deterministic answer. In many settings, multiple answers exist for a question. Thus, it would be interesting to extend TRM to generative tasks.

Acknowledgements

Thank you Emy Gervais for your invaluable support and extra push. This research was enabled in part by computing resources, software, and technical assistance provided by Mila and the Digital Research Alliance of Canada.

References

ARC Prize Foundation. The Hidden Drivers of HRM's Performance on ARC-AGI. https://arcprize.org/blog/hrm-analysis, 2025a. [Online; accessed 2025-09-15].

ARC Prize Foundation. ARC-AGI Leaderboard. https://arcprize.org/leaderboard, 2025b. [Online; accessed 2025-09-24].

Bai, S., Kolter, J. Z., and Koltun, V. Deep equilibrium models. *Advances in neural information processing systems*, 32, 2019.

Bai, X. and Melas-Kyriazi, L. Fixed point diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9430–9440, 2024.

Brock, A., Donahue, J., and Simonyan, K. Large scale gan training for high fidelity natural image synthesis. *arXiv preprint arXiv:1809.11096*, 2018.

Chollet, F. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.

Chollet, F., Knoop, M., Kamradt, G., Landers, B., and Pinkard, H. Arc-agi-2: A new challenge for frontier ai reasoning systems. *arXiv preprint arXiv:2505.11831*, 2025.

Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H. W., Sutton, C., Gehrmann, S., et al. Palm: Scaling language modeling with pathways. *Journal of Machine Learning Research*, 24(240):1–113, 2023.

Dillion, T. Tdoku: A fast sudoku solver and generator. https://t-dillon.github.io/tdoku/, 2025.

Elman, J. L. Finding structure in time. *Cognitive science*, 14(2):179–211, 1990.

Fedus, W., Zoph, B., and Shazeer, N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

Geng, Z. and Kolter, J. Z. Torchdeq: A library for deep equilibrium models. *arXiv preprint arXiv:2310.18605*, 2023.

Hendrycks, D. and Gimpel, K. Gaussian error linear units (gelus). *arXiv preprint arXiv:1606.08415*, 2016.

Jang, Y., Kim, D., and Ahn, S. Hierarchical graph generation with k2-trees. In *ICML 2023 Workshop on Structured Probabilistic Inference Generative Modeling*, 2023.

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., and Amodei, D. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Kingma, D. P. and Ba, J. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

Krantz, S. G. and Parks, H. R. *The implicit function theorem: history, theory, and applications*. Springer Science & Business Media, 2002.

LeCun, Y. Une procedure d'apprentissage ponr reseau a seuil asymetrique. *Proceedings of cognitiva 85*, pp. 599–604, 1985.

Lehnert, L., Sukhbaatar, S., Su, D., Zheng, Q., Mcvay, P., Rabbat, M., and Tian, Y. Beyond a*: Better planning with transformers via search dynamics bootstrapping. *arXiv preprint arXiv:2402.14083*, 2024.

Lillicrap, T. P. and Santoro, A. Backpropagation through time and the brain. *Current opinion in neurobiology*, 55:82–89, 2019.

Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

Moskvichev, A., Odouard, V. V., and Mitchell, M. The conceptarc benchmark: Evaluating understanding and generalization in the arc domain. *arXiv preprint arXiv:2305.07141*, 2023.

Palm, R., Paquet, U., and Winther, O. Recurrent relational networks. *Advances in neural information processing systems*, 31, 2018.

Park, K. Can convolutional neural networks crack sudoku puzzles? https://github.com/Kyubyong/sudoku, 2018.

Prieto, L., Barsbey, M., Mediano, P. A., and Birdal, T. Grokking at the edge of numerical stability. *arXiv preprint arXiv:2501.04697*, 2025.

Rumelhart, D. E., Hinton, G. E., and Williams, R. J. Learning internal representations by error propagation. Technical report, 1985.

Shazeer, N. Glu variants improve transformer. *arXiv preprint arXiv:2002.05202*, 2020.

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., and Dean, J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

Snell, C., Lee, J., Xu, K., and Kumar, A. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.

Song, Y. and Ermon, S. Improved techniques for training score-based generative models. *Advances in neural information processing systems*, 33:12438–12448, 2020.

Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., and Liu, Y. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.

Tolstikhin, I. O., Houlsby, N., Kolesnikov, A., Beyer, L., Zhai, X., Unterthiner, T., Yung, J., Steiner, A., Keysers, D., Uszkoreit, J., et al. Mlp-mixer: An all-mlp architecture for vision. *Advances in neural information processing systems*, 34:24261–24272, 2021.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

Wang, G., Li, J., Sun, Y., Chen, X., Liu, C., Wu, Y., Lu, M., Song, S., and Yadkori, Y. A. Hierarchical reasoning model. *arXiv preprint arXiv:2506.21734*, 2025.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Werbos, P. Beyond regression: New tools for prediction and analysis in the behavioral sciences. *PhD thesis, Committee on Applied Mathematics, Harvard University, Cambridge, MA*, 1974.

Werbos, P. J. Generalization of backpropagation with application to a recurrent gas market model. *Neural networks*, 1(4):339–356, 1988.

Zhang, B. and Sennrich, R. Root mean square layer normalization. *Advances in Neural Information Processing Systems*, 32, 2019.

Hyper-parameters and setup

All models are trained with the AdamW optimizer(Loshchilov & Hutter, 2017; Kingma & Ba, 2014) with $\beta_1 = 0.9$, $\beta_2 = 0.95$, small learning rate warm-up (2K iterations), batch-size 768, hidden-size of 512, $N_{sup} = 16$ max supervision steps, and stable-max loss (Prieto et al., 2025) for improved stability. TRM uses an Exponential Moving Average (EMA) of 0.999. HRM uses $n = 2, T = 2$ with two 4-layers networks, while we use $n = 6, T = 3$ with one 2-layer network.

For Sudoku-Extreme and Maze-Hard, we train for 60k epochs with learning rate 1e-4 and weight decay 1.0. For ARC-AGI, we train for 100K epochs with learning rate 1e-4 (with 1e-2 learning rate for the embeddings) and weight decay 0.1. The numbers for Deepseek R1, Claude 3.7 8K, O3-mini-high, Direct prediction, and HRM from the Table 4 and 5 are taken from Wang et al. (2025). Both HRM and TRM add an embedding of shape $[0, 1, D]$ on Sudoku-Extreme and Maze-Hard to the input. For ARC-AGI, each puzzle (containing 2-3 training examples and 1-2 test examples) at each data-augmentation is given a specific embedding of shape $[0, 1, D]$ and, at test-time, the most common answer out of the 1000 data augmentations is given as answer.

Experiments on Sudoku-Extreme were ran with 1 L40S with 40Gb of RAM for generally less than 36 hours. Experiments on Maze-Hard were ran with 4 L40S with 40Gb of RAM for less than 24 hours. Experiments on ARC-AGI were ran for around 3 days with 4 H100 with 80Gb of RAM.

Ideas that failed

In this section, we quickly mention a few ideas that did not work to prevent others from making the same mistake.

We tried replacing the SwiGLU MLPs by SwiGLU Mixture-of-Experts (MoEs) (Shazeer et al., 2017; Fedus et al., 2022), but we found generalization to decrease massively. MoEs clearly add too much unnecessary capacity, just like increasing the number of layers does.

Instead of back-propagating through the whole $n + 1$ recursions, we tried a compromise between HRM 1-step gradient approximation, which back-propagates through the last 2 recursions. We did so by decoupling n from the *number of last recursions* k that we back-propagate through. For example, while $n = 6$ requires 7 steps with gradients in TRM, we can use gradients for only the $k = 4$ last steps. However, we found that this did not help generalization in any way, and it made the approach more complicated. Back-propagating through the whole $n + 1$ recursions makes the most sense and works best.

We tried removing ACT with the option of stopping when the solution is reached, but we found that generalization dropped significantly. This can probably be attributed to the fact that the model is spending too much time on the same data samples rather than focusing on learning on a wide range of data samples.

We tried weight tying the input embedding and output head, but this was too constraining and led to a massive generalization drop.

We tried using TorchDEQ (Geng & Kolter, 2023) to replace the recursion steps by fixed-point iteration as done by Deep Equilibrium Models (Bai et al., 2019). This would provide a better justification for the 1-step gradient approximation. However, this slowed down training due to the fixed-point iteration and led to worse generalization. This highlights the fact that converging to a fixed-point is not essential.

Algorithms with different number of latent features

```
def latent_recursion(x, z, n=6):
    for i in range(n+1): # latent recursion
        z = net(x, z)
    return z

def deep_recursion(x, z, n=6, T=3):
    # recursing T−1 times to improve z (no gradients needed)
    with torch.no_grad():
        for j in range(T-1):
            z = latent_recursion(x, z, n)
    # recursing once to improve z
    z = latent_recursion(x, z, n)
    return z.detach(), output_head(y), Q_head(y)

# Deep Supervision
for x_input, y_true in train_dataloader:
    z = z_init
    for step in range(N_supervision):
        x = input_embedding(x_input)
        z, y_hat, q_hat = deep_recursion(x, z)
        loss = softmax_cross_entropy(y_hat, y_true)
        loss += binary_cross_entropy(q_hat, (y_hat == y_true))
        z = z.detach()
        loss.backward()
        opt.step()
        opt.zero_grad()
        if q[0] > 0: # early−stopping
            break
```

Figure 4. Pseudocode of TRM using a single-z with deep supervision training in PyTorch.

```
def latent_recursion(x, y, z, n=6):
    for i in range(n): # latent recursion
        z[i] = net(x, y, z[0], ... , z[n−1])
    y = net(y, z[0], ... , z[n−1]) # refine output answer
    return y, z

def deep_recursion(x, y, z, n=6, T=3):
    # recursing T−1 times to improve y and z (no gradients needed)
    with torch.no_grad():
        for j in range(T-1):
            y, z = latent_recursion(x, y, z, n)
    # recursing once to improve y and z
    y, z = latent_recursion(x, y, z, n)
    return (y.detach(), z.detach()), output_head(y), Q_head(y)

# Deep Supervision
for x_input, y_true in train_dataloader:
    y, z = y_init, z_init
    for step in range(N_supervision):
        x = input_embedding(x_input)
        (y, z), y_hat, q_hat = deep_recursion(x, y, z)
        loss = softmax_cross_entropy(y_hat, y_true)
        loss += binary_cross_entropy(q_hat, (y_hat == y_true))
        loss.backward()
        opt.step()
        opt.zero_grad()
        if q[0] > 0: # early−stopping
            break
```

Figure 5. Pseudocode of TRM using multi-scale z with deep supervision training in PyTorch.

Example on Sudoku-Extreme

						8	3	1
	9			6	8		7	
			3		5			
	6	8						
					6			2
7	4							3
					9			4
2				4			1	
6				2			5	7

Input x

5	2	6	7	9	4	8	3	1
3	9	1	2	6	8	4	7	5
4	8	7	3	1	5	2	9	6
1	6	8	5	3	2	7	4	9
9	3	5	4	7	6	1	8	2
7	4	2	9	8	1	5	6	3
8	7	3	1	5	9	6	2	4
2	5	9	6	4	7	3	1	8
6	1	4	8	5	3	9	5	7

Output y

5	2	6	7	9	4	8	3	1
3	9	1	2	6	8	4	7	5
4	8	7	3	1	5	2	9	6
1	6	8	5	3	2	7	4	9
9	3	5	4	7	6	1	8	2
7	4	2	9	8	1	5	6	3
8	7	3	1	5	9	6	2	4
2	5	9	6	4	7	3	1	8
6	1	4	8	5	3	9	5	7

Tokenized z_H (denoted y in TRM)

5		5	4	9	4		6	3
4		3	1			4	6	5
4	8	4		3		6	6	4
9		6	5	3		5	4	
	3	5	4	3		5	4	4
6		3		3	3	5	8	8
3	3	3	6	5		6	6	4
7	5		6		3	3	6	6
4	3	4	8		3	6	6	4

Tokenized z_L (denoted z in TRM)

Figure 6. This Sudoku-Extreme example shows an input, expected output, and the tokenized z_H and z_L (after reversing the embedding and using argmax) for a pretrained model. This highlights the fact that z_H corresponds to the predicted response, while z_L is a latent feature that cannot be decoded to a sensible output unless transformed into z_H by f_H.